



# Using CRSPR-Cas9 for Treatment of Alzheimer's Disease

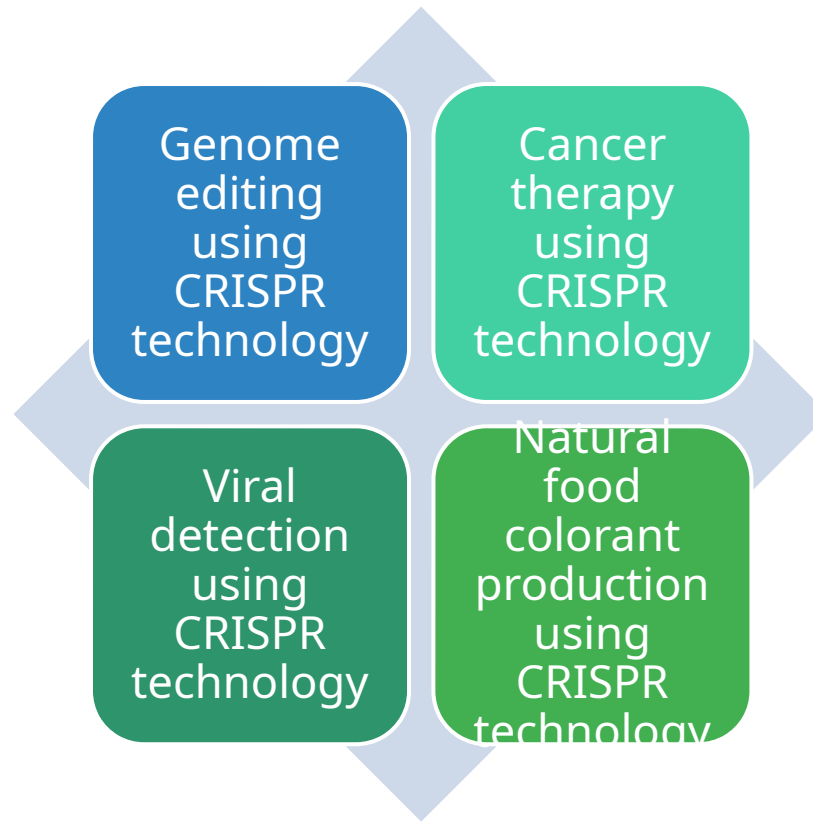
Safiy Khan

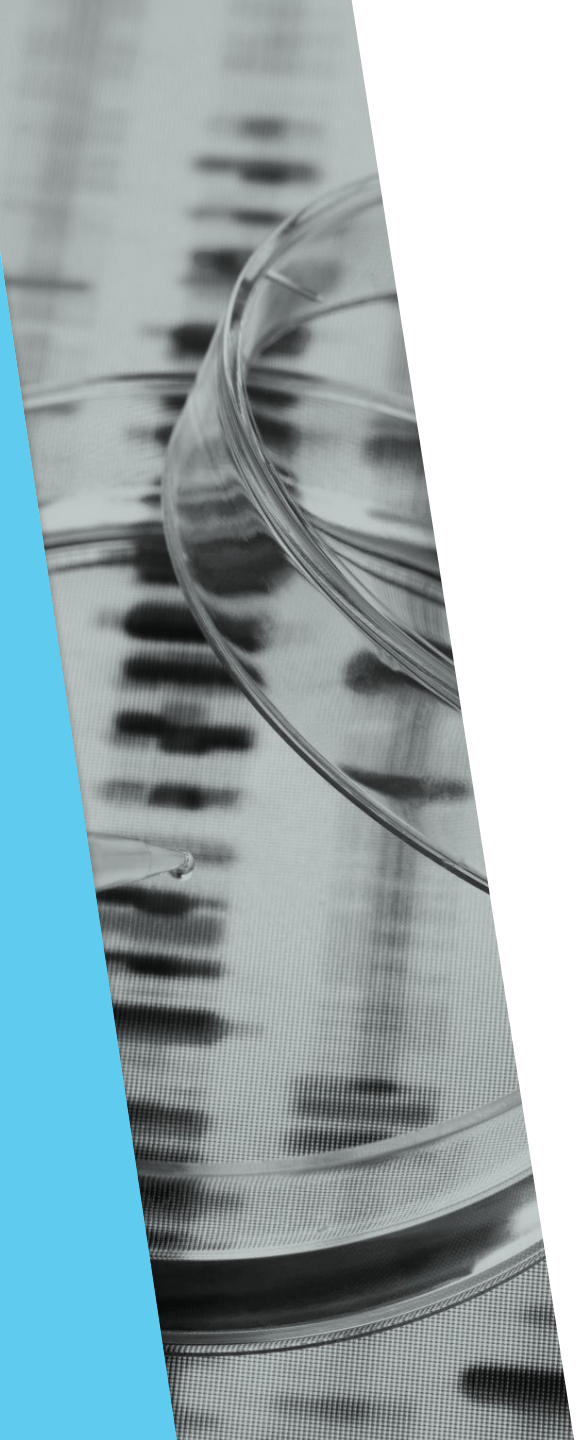


# Introduction to CRISPR technology

- CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) technology
- Revolutionary technology in the field of biotechnology
- Accurate and efficient method for targeted genome editing
- Opened up new horizons in the fields of genetics, genomics, medicine, agriculture, and environmental protection

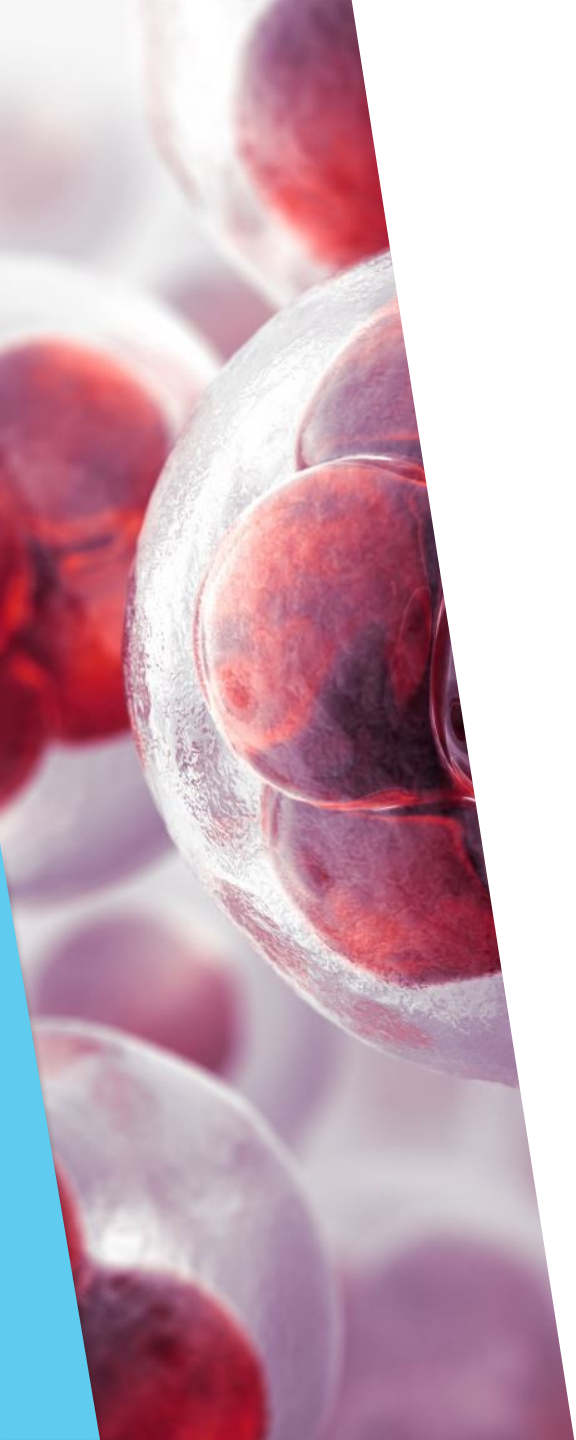
# Recent advancements in CRISPR technology



A vertical strip on the left side of the slide showing a grayscale image of a gel electrophoresis result, with multiple horizontal bands of varying intensity, suggesting DNA sequencing or analysis.

# Genome editing analysis using the CRISPR detector

- Pipeline for detecting, visualizing, and annotating genome-wide mutations induced by genome editing events
  - Analyze datasets generated from whole-genome sequencing (WGS) with improved accuracy
  - Integrated structural variation calling and includes functional and clinical annotation of editing-induced mutations
  - Enables rapid and efficient detection of mutations induced by genome editing events
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- A decorative background on the right side of the slide consisting of overlapping, semi-transparent blue geometric shapes, primarily triangles and polygons, creating a modern, abstract design.

A vertical strip on the left side of the slide showing a microscopic view of cells, likely cancer cells, with a reddish-pink hue. The cells are clustered and have a textured, almost crystalline appearance.

# CRISPR technology in cancer research

- Understanding and targeting drug resistance mechanisms and factors using CRISPR technology
  - Modeling, screening, and reversing drug resistance in cancer using CRISPR technology
  - Developing CRISPR-based therapeutic approaches to sensitize cancer cells to chemotherapy
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- A decorative background on the right side of the slide consisting of overlapping, semi-transparent blue geometric shapes, primarily triangles and polygons, creating a modern, abstract design.

A composite image featuring a petri dish in the foreground, a pipette tip on the left, and a background of a DNA gel electrophoresis image with multiple lanes of colored bands.

## CRISPR-based genome editing for disease treatment and diagnosis

- Promising method for the treatment of genetic diseases
- Correcting the mutation responsible for sickle cell anemia using a CRISPR-Cas9 system
- Correction with high efficiency, and the corrected cells were able to produce normal hemoglobin
- Automated CRISPR-based diagnostic tool demonstrated comparable performance to the routine RT-PCR testing for diagnosing COVID-19

# CRISPR technology in miscellaneous industries



Early diagnosis of RNA virus in aquaculture using CRISPR-Cas13a assay



Directly detecting Taura syndrome virus (TSV) from shrimp RNA



Improving natural food colorant production using CRISPR-based engineering in *Monascus ruber*

# Proposal Introduction

- ▶ Proposed research on treatment for Alzheimer's disease
- ▶ Alzheimer's disease is a neurodegenerative disorder
  - ▶ Affects over 5 million people in the United States alone
  - ▶ Currently no cure for this disease.
- ▶ Develop a novel therapeutic approach to treating Alzheimer's using CRISPR-based gene editing.

# Understanding Alzheimer's Disease

- ▶ Progressive neurodegenerative disorder
- ▶ Affects the brain's ability to function properly
- ▶ Characterized by the accumulation of beta-amyloid plaques and tau tangles in the brain
  - ▶ Lead to:
    - ▶ The death of brain cells
    - ▶ The progressive loss of cognitive function.
- ▶ Exact cause unknown, but genetic factors may play a significant role in development of disease.

# CRISPR-Based Gene Editing for Alzheimer's Disease

- ▶ Use CRISPR-based gene editing to target the genes responsible for the accumulation of beta-amyloid plaques and tau tangles in the brain.
- ▶ Aim to prevent the formation of these toxic proteins
  - ▶ Halt the progression of Alzheimer's disease.

# Challenges of CRISPR-Based Gene Editing for Alzheimer's Disease

- ▶ Extremely promising technology, but not without challenges:
  - ▶ Key challenge: delivering the CRISPR system to the brain
    - ▶ Requires crossing the blood-brain barrier.
  - ▶ Specificity and safety of the CRISPR system
    - ▶ Crucial in preventing off-target effects/unintended consequences.

# Delivery Methods for CRISPR-Based Gene Editing

- ▶ Proposed delivery methods to the brain
  - ▶ Viral vectors
  - ▶ Nanoparticles
  - ▶ Exosomes.
- ▶ In-depth research needed to determine most effective delivery method.

# Specificity and Safety of CRISPR-Based Gene Editing

Problem: specificity and safety

Potential solutions:

- Use of modified version of the CRISPR system
  - Designed to be more specific and accurate
  - Example: CRISPR-Cas13d system
- Use inducible promoters
  - Greater control over the timing and duration of gene expression
    - Reduces the risk of off-target effects.

# Animal Studies for CRISPR-Based Gene Editing in Alzheimer's Disease

- ▶ Testing of efficacy and safety of approach
- ▶ Conduct animal studies using a mouse model of Alzheimer's disease:
  - ▶ The mice receive CRISPR-based gene editing therapy
  - ▶ Monitor cognitive function
  - ▶ Assess impact of treatment on accumulation of beta-amyloid plaques and tau tangles

# Future Directions for CRISPR-Based Gene Editing in Alzheimer's Disease

- ▶ If successful, research could usher in development of new therapeutic approach for Alzheimer's disease.
- ▶ If successful, future directions include:
  - ▶ Optimize CRISPR system for greater specificity and safety
  - ▶ Developing effective delivery methods
  - ▶ Conducting clinical trials to test the safety and efficacy of method